

output is proportional to the input. For multiple-input linear circuits, the superposition principle states that any output can be found by summing the output produced when each input acts alone.

- A Thévenin equivalent circuit consists of a voltage source in series with a resistance. A Norton equivalent circuit consists of a current source in parallel with a resistance. The Thévenin and Norton equivalent circuits are related by a source transformation.
- The parameters of the Thévenin and Norton equivalent circuits can be determined using any two of the following: (1) the open-circuit voltage at the interface, (2) the short-circuit current at the interface, and (3) the equivalent resistance of the source circuit with all sources turned off.
- The parameters of the Thévenin and Norton equivalent circuits can also be determined using circuit reduction methods or by directly solving for the source i - v relationship using node-voltage or mesh-current analysis.
- For a fixed source and an adjustable load, the maximum interface signal levels are $v_{\text{MAX}} = v_{\text{OC}}$ ($R_L = \infty$), $i_{\text{MAX}} = i_{\text{SC}}$ ($R_L = 0$), and $p_{\text{MAX}} = v_{\text{OC}} i_{\text{SC}} / 4$ ($R_L = R_T$). When $R_L = R_T$, the source and load are said to be matched.
- Interface signal transfer conditions are specified in terms of the voltage, current, or power delivered to the load. The design constraints depend on the signal conditions specified and the circuit parameters that are adjustable. Some design requirements may require a two-port interface circuit. An interface design problem may have one, many, or no solutions.

PROBLEMS

OBJECTIVE 3-1 GENERAL CIRCUIT ANALYSIS (SECT. 3-1 TO 3-2)

Given a circuit consisting of linear resistors and independent sources,

- (Formulation) Write node-voltage or mesh-current equations for the circuit.
- (Solution) Solve the equations from (a) for selected signal variables or input-output relationships using classical or software computational techniques.

Node-voltage method: See Examples 3-1, 3-2, 3-3, 3-4, 3-5, 3-6, 3-7 and Exercises 3-2, 3-3, 3-4, 3-5, 3-6, 3-7, 3-8, 3-9, 3-10, 3-11, 3-12, 3-13.

Mesh-current method: See Examples 3-8, 3-9, 3-10 and Exercises 3-14, 3-15, 3-16, 3-17, 3-18, 3-19, 3-20.

- 3-1** (a) Formulate node-voltage equations for the circuit in Figure P3-1. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

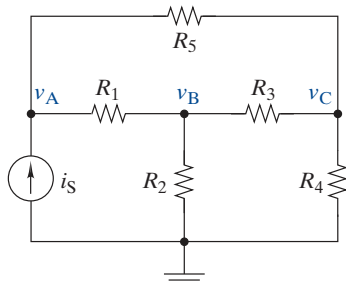


FIGURE P3-1

- 3-2** (a) Formulate node-voltage equations for the circuit in Figure P3-2. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Solve these equations for v_A and v_B .

(c) Use these results to find v_x and i_x .

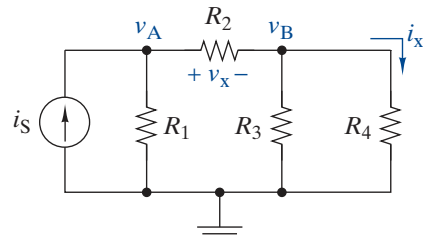


FIGURE P3-2

- 3-3** (a) Formulate node-voltage equations for the circuit in Figure P3-3. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Solve these equations for v_A and v_B .

(c) Use these results to find v_x and i_x .

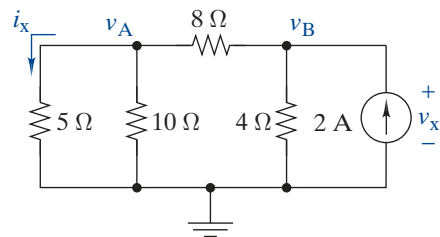


FIGURE P3-3

- 3-4 (a)** Formulate node-voltage equations for the circuit in Figure P3-4.
(b) Solve these equations for v_A and v_B .
(c) Use these results to find v_x and i_x .

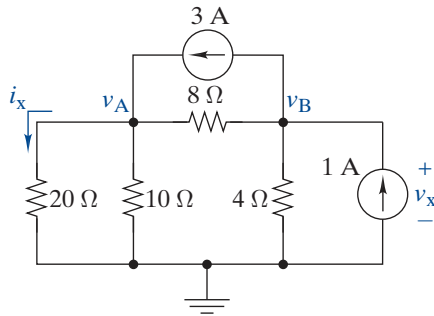


FIGURE P3-4

- 3-5 (a)** Formulate node-voltage equations for the circuit in Figure P3-5. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve these equations for v_A and v_C .
(c) Use these results to find v_x and i_x .

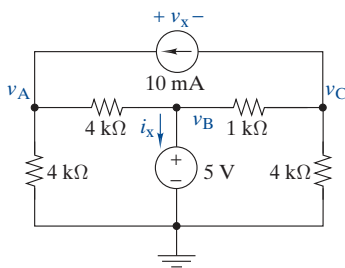


FIGURE P3-5

- 3-6 (a)** Choose a ground wisely and formulate node-voltage equations for the circuit in Figure P3-6.
(b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$, $v_s = 25 \text{ V}$, and $i_s = 1 \text{ mA}$.

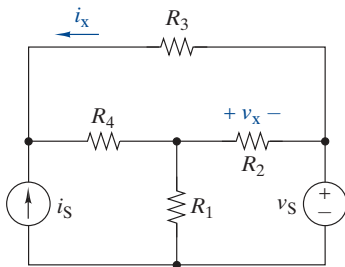


FIGURE P3-6

- 3-7** The following are a set of node-voltage equations; draw the circuit they represent.

$$\begin{aligned} v_B - v_A &= v_s \\ \frac{v_B - v_A}{R_1} + \frac{v_B - v_C}{R_2} - i_s &= 0 \\ \frac{v_C - v_A}{R_3} + \frac{v_C - v_B}{R_2} + \frac{v_C}{R_4} &= 0 \\ v_D &= 0 \end{aligned}$$

- 3-8 (a)** Choose a ground wisely and formulate node-voltage equations for the circuit in Figure P3-8.
(b) Solve for v_x and i_x .

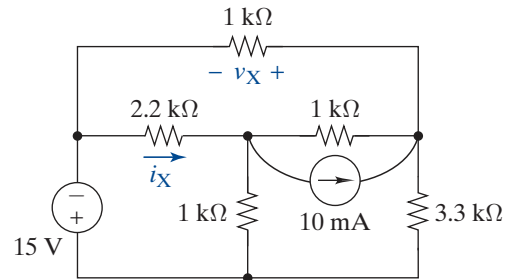


FIGURE P3-8

- 3-9 (a)** Formulate node-voltage equations for the circuit in Figure P3-9.
(b) Use MATLAB to find symbolic expressions for the node voltages in terms of the parameters in the circuit.
(c) Find numeric values for v_A , v_B , and v_C when $R_1 = 1.5 \text{ k}\Omega$, $R_2 = 2.2 \text{ k}\Omega$, $R_3 = 3.3 \text{ k}\Omega$, $R_4 = 4.7 \text{ k}\Omega$, and $i_{S1} = i_{S2} = 2 \text{ mA}$.
(d) Use OrCAD to verify your solution to part (c) is correct.

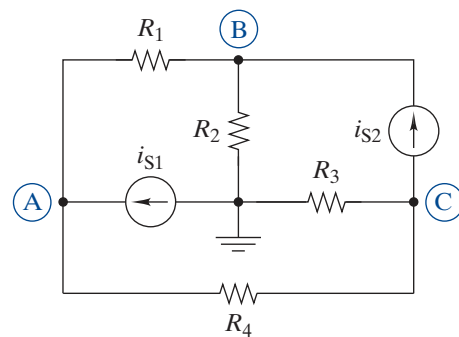


FIGURE P3-9

- 3-10(a)** Formulate node-voltage equations for the circuit in Figure P3-10.
(b) Solve for v_x and i_x when $R_1 = R_4 = 2 \text{ k}\Omega$, $R_2 = R_3 = 500 \Omega$, $R_x = 750 \Omega$, and $v_s = 15 \text{ V}$.
(c) Repeat (b) when R_4 is a variable resistor that varies from 10Ω to $10,000 \Omega$. At what value of R_4 is the voltage across $R_x = 0 \text{ V}$? *Hint:* Use OrCAD to vary R_4 by trial and error to approach the answer.

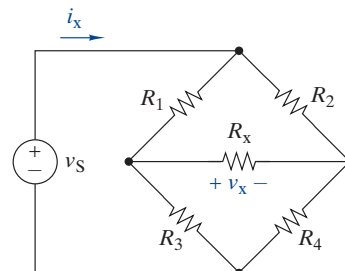


FIGURE P3-10

- 3-11(a)** Formulate node-voltage equations for the circuit in Figure P3-11.
(b) Solve for v_x and i_x when $R_1 = R_4 = 2 \text{ k}\Omega$, $R_2 = R_3 = 500 \text{ }\Omega$, $R_x = 750 \text{ }\Omega$, and $v_s = 15 \text{ V}$.

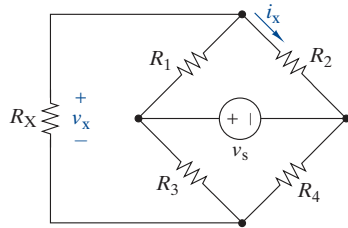


FIGURE P3-11

- 3-12(a)** Formulate node-voltage equations for the circuit in Figure P3-12. (*Hint: use a supernode.*)
(b) Solve for v_x and i_x .

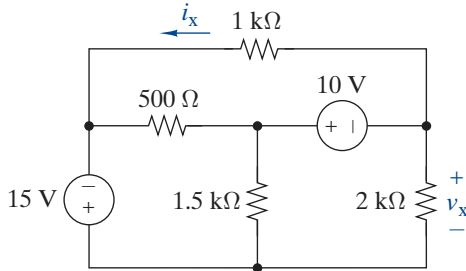


FIGURE P3-12

- 3-13(a)** Formulate mesh-current equations for the circuit in Figure P3-13. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve for i_A and i_B .
(c) Use these results to find v_x and i_x .

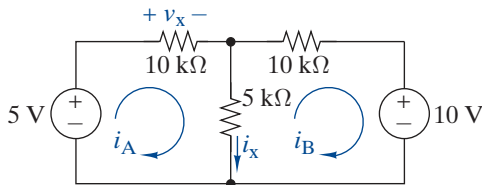


FIGURE P3-13

- 3-14(a)** Formulate mesh-current equations for the circuit in Figure P3-14. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve for i_A , i_B and i_C .
(c) Use these results to find v_x and i_x .

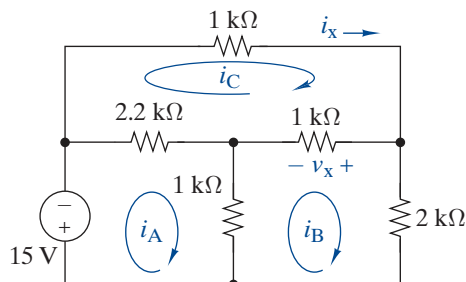


FIGURE P3-14

- 3-15(a)** Formulate mesh-current equations for the circuit in Figure P3-15. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve for i_A and i_B .
(c) Use these results to find v_x and i_x .

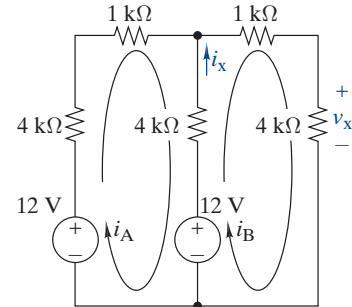


FIGURE P3-15

- 3-16(a)** Formulate mesh-current equations for the circuit in Figure P3-16. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve for i_A and i_B .
(c) Use these results to find v_x and i_x .

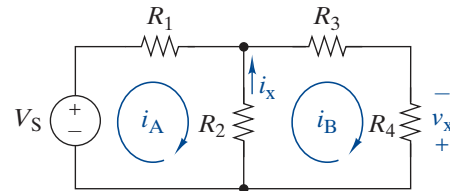


FIGURE P3-16

- 3-17(a)** Formulate mesh-current equations for the circuit in Figure P3-17.
(b) Formulate node-voltage equations for the circuit in Figure P3-17.
(c) Which set of equations would be easier to solve? Why?
(d) Using MATLAB, find v_x and i_x in terms of the mesh-current variables.
(e) Using MATLAB, find v_x and i_x in terms of the node-voltage variables.

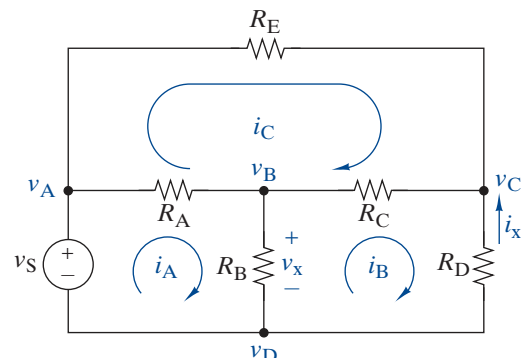


FIGURE P3-17

- 3-18(a) Formulate mesh-current equations for the circuit in Figure P3-18.
 (b) Solve for v_x and i_x when $R_1 = 2 \text{ k}\Omega$, $R_2 = 3 \text{ k}\Omega$, $R_3 = 500 \Omega$, $R_4 = 2.5 \text{ k}\Omega$, $R_5 = 2 \text{ k}\Omega$, $i_s = 10 \text{ mA}$, and $v_s = 24 \text{ V}$.
 (c) Find the total power dissipated in the circuit.

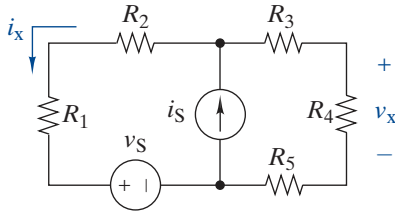


FIGURE P3-18

- 3-19(a) For the circuit of Figure P3-19 solve for i_A , i_B and i_C using two supermesh equations.
 (b) Use these results to find v_x .

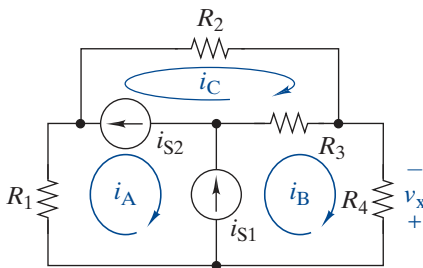


FIGURE P3-19

- 3-20(a) Formulate mesh-current equations for the circuit in Figure P3-20.
 (b) Use MATLAB to find symbolic expressions for v_x and i_x in terms of the parameters in the circuit.
 (c) Find numeric values for v_x and i_x when $R_1 = R_2 = 5.6 \text{ k}\Omega$, $R_3 = 2.2 \text{ k}\Omega$, $R_4 = 1.5 \text{ k}\Omega$, $i_s = 3 \text{ mA}$, $v_{s1} = 15 \text{ V}$, and $v_{s2} = 5 \text{ V}$.
 (d) Find the power supplied by v_{s1} .
 (e) Use OrCAD to verify your solutions to parts (c) and (d) are correct.

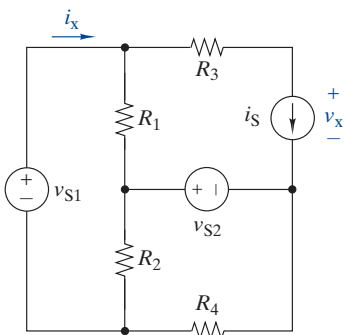


FIGURE P3-20

- 3-21 The circuit in Figure P3-21 seems to require two supermeshes since both current sources appear in two meshes. However, sometimes rearranging the circuit diagram will eliminate the need for a supermesh.
 (a) Show that supermeshes can be avoided in Figure P3-21 by rearranging the connection of resistor R_6 .
 (b) Formulate mesh-current equations for the modified circuit as redrawn in (a).
 (c) Solve for v_x when $R_1 = R_2 = R_3 = R_4 = 4 \text{ k}\Omega$, $R_5 = R_6 = 2 \text{ k}\Omega$, $i_{s1} = 80 \text{ mA}$, and $i_{s2} = 40 \text{ mA}$.

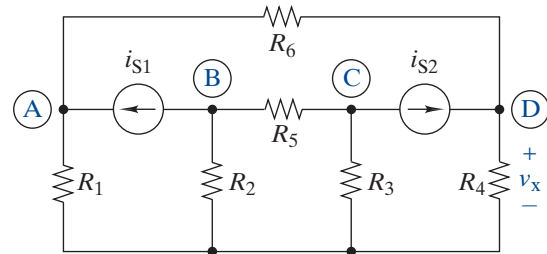


FIGURE P3-21

- 3-22(a) Formulate mesh-current equations for the circuit in Figure P3-22.
 (b) Formulate node-voltage equations for the circuit in Figure P3-22.
 (c) Which set of equations would be easier to solve? Why?
 (d) Find v_x and i_x using whichever method you prefer.

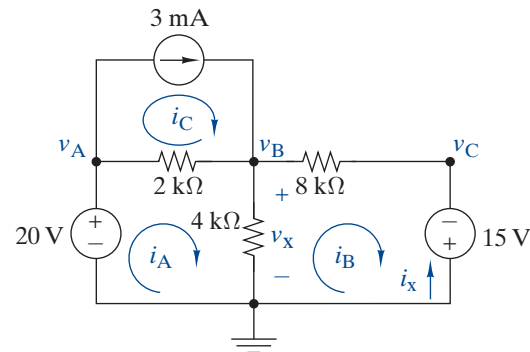


FIGURE P3-22

- 3-23 Use mesh-current, node-voltage or simple engineering intuition to find the input resistance of the circuit in Figure P3-23.

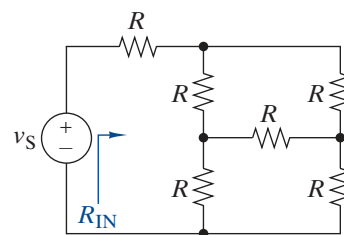


FIGURE P3-23

3–24 In Figure P3–24 all of the resistors are $1\text{ k}\Omega$ and $v_S = 10\text{ V}$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

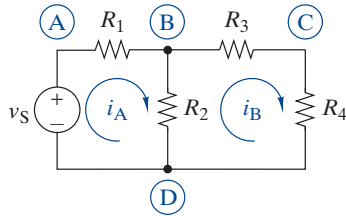


FIGURE P3–24

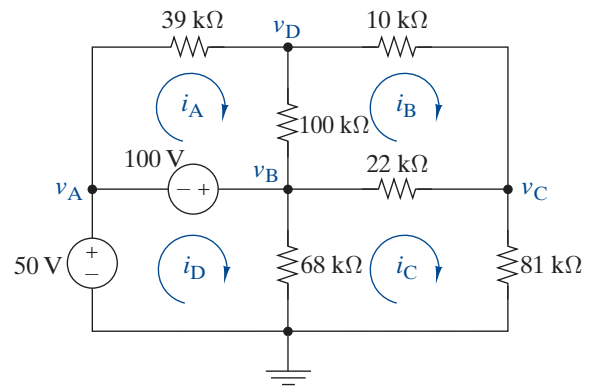


FIGURE P3–27

3–25 Use Figure P3–24 and MATLAB to solve the following problems:

- Using mesh-current analysis, find a symbolic expression for i_A in terms of the circuit parameters.
- Compute the ratio i_A/v_S .
- Find a symbolic expression for the equivalent resistance of the circuit by combining resistors in series and parallel. Compare your answer to the results from part (b).

3–26(a) Formulate mesh-current equations for the circuit in Figure P3–26.

(b) Formulate node-voltage equations for the circuit in Figure P3–26.

(c) Which set of equations would be easier to solve? Why?

(d) Use OrCAD to find the node voltages v_A and v_B and the mesh-currents i_A , i_B and i_C in Figure P3–26.

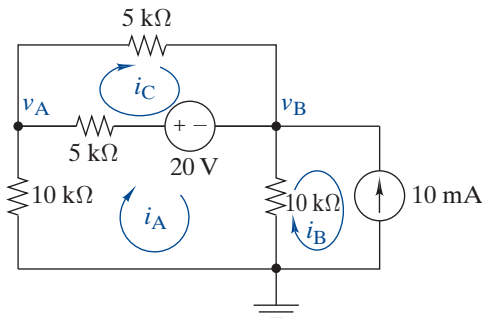


FIGURE P3–26

3–27(a) Formulate mesh-current equations for the circuit in Figure P3–27. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Use MATLAB and mesh-current analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3–27.

(c) Formulate node-voltage equations for the circuit in Figure P3–27. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(d) Use MATLAB and node-voltage analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3–27 and compare the effort required with each technique, i.e. mesh-current versus node-voltage.

(e) Use OrCAD to verify your results in the previous two parts.

OBJECTIVE 3–2 LINEARITY PROPERTIES (SECT. 3–3)

(a) Given a circuit containing linear resistors and one independent source, use the proportionality principle to find selected signal variables.

(b) Given a circuit containing linear resistors and two or more independent sources, use the superposition principle to find selected signal variables.

See Examples 3–11, 3–12, 3–13 and Exercises 3–21, 3–22, 3–23, 3–24 for proportionality.

See Examples 3–14, 3–15 and Exercises 3–25, 3–26, 2–27 for superposition.

3–28 Find the proportionality constant $K = v_O/v_S$ for the circuit in Figure P3–28.

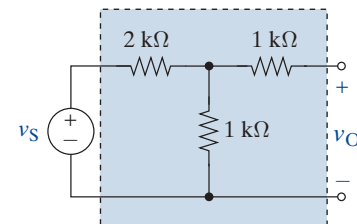


FIGURE P3–28

3–29 Find the proportionality constant $K = i_O/v_S$ for the circuit in Figure P3–29.

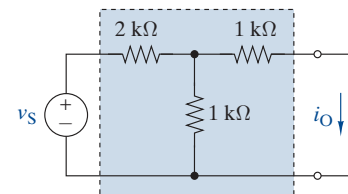


FIGURE P3–29

- 3–30 Find the proportionality constant $K = v_O/i_S$ for the circuit in Figure P3–30.

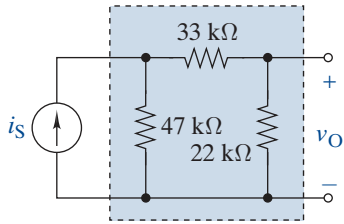


FIGURE P3–30

- 3–31 Find the proportionality constant $K = i_O/i_S$ for the circuit in Figure P3–31.

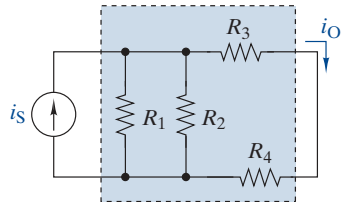


FIGURE P3–31

- 3–32 Find the proportionality constant $K = v_O/v_S$ for the circuit in Figure P3–32.

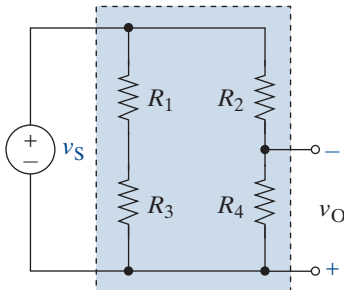


FIGURE P3–32

- 3–33 Use the unit output method to find K and v_O in Figure P3–33.

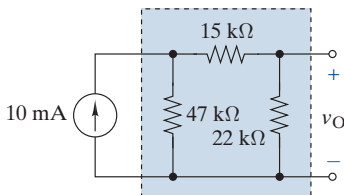


FIGURE P3–33

- 3–34 Use the unit output method to find K and v_O in Figure P3–34.

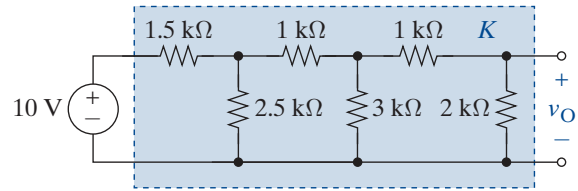


FIGURE P3–34

- 3–35 Use the unit output method to find K and i_O in Figure P3–35.

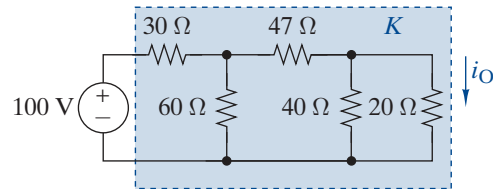


FIGURE P3–35

- 3–36 Use the superposition principle to find v_O in Figure P3–36.

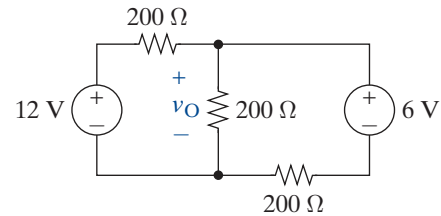


FIGURE P3–36

- 3–37 Use the superposition principle to find i_O in Figure P3–37. Verify your answer using OrCAD.

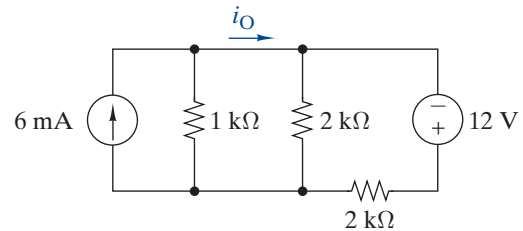


FIGURE P3–37

- 3–38 Use the superposition principle to find v_O in Figure P3–38.

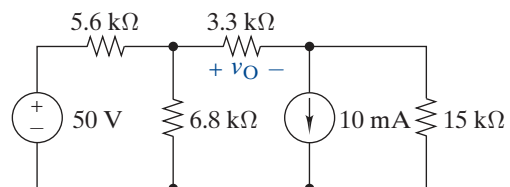


FIGURE P3–38

3-39 Use the superposition principle to find v_O in Figure P3-39.

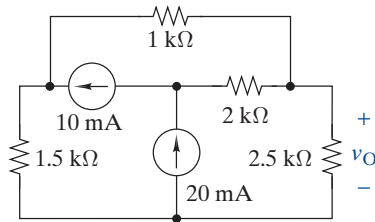


FIGURE P3-39

3-40 Use the superposition principle to find i_O in Figure P3-40. Verify your answer using OrCAD.

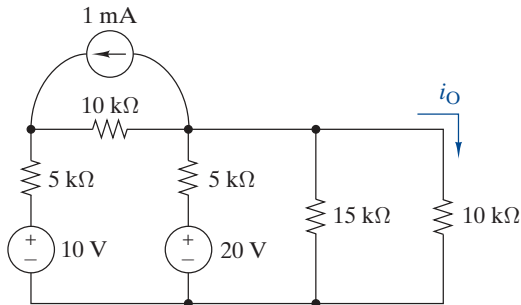


FIGURE P3-40

3-41 (a) Use the superposition principle to find v_O in terms of v_1 , v_2 , and R in Figure P3-41 (This circuit is a 2-bit R - $2R$ network)
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

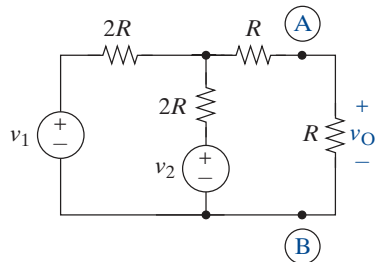


FIGURE P3-41

3-42(a) Use the superposition principle to find v_O in terms of v_S , i_S , and R in Figure P3-42.
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

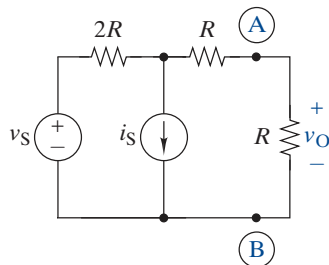


FIGURE P3-42

3-43 A linear circuit containing two sources drives a 100-Ω load resistor. Source number 1 delivers 1 W to the load when source

number 2 is off. Source number 2 delivers 4 W to the load when source number 1 is off. Find the power delivered to the load when both sources are on. *Hint:* The answer is not 5 W. Why?

3-44 A linear circuit is driven by an independent voltage source $v_S = 10$ V and an independent current source $i_S = 10$ mA. The output voltage is $v_O = 2$ V when the voltage source is on and the current source off. The output is $v_O = 1$ V when both sources are on. Find the output voltage when $v_S = 20$ V and $i_S = -20$ mA.

3-45 A certain linear circuit has four input voltages and one output voltage v_O . The following table lists the output for different values of the four inputs. Find the input-output relationship for the circuit. Specifically, find an expression for v_O in terms of the four input voltages.

$v_{S1}(V)$	$v_{S2}(V)$	$v_{S3}(V)$	$v_{S4}(V)$	$v_O(V)$
2	4	-4	1	20
1	2	2	1.5	-4
1	4	2	2	-1
0	5	3	-1	3

3-46 When the current source is turned off in the circuit of Figure P3-46 the voltage source delivers 25 W to the load. How much power does it deliver to the load when both sources are on? Explain your answer.

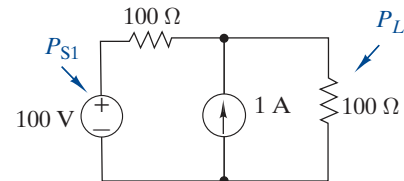


FIGURE P3-46

OBJECTIVE 3-3 THÉVENIN AND NORTON EQUIVALENT CIRCUITS (SECT. 3-4)

Given a circuit containing linear resistors and independent sources,
(a) Find the Thévenin or Norton equivalent at a specified pair of terminals.

(b) Use the Thévenin or Norton equivalent to find the signals delivered to linear or nonlinear loads.

See Examples 3-16, 3-17, 3-18, 3-19, 3-20, 3-21 and Exercises 3-28, 3-29, 3-30, 3-31, 3-32.

3-47 For the circuit in Figure P3-47 find its Thévenin and Norton equivalent circuits.

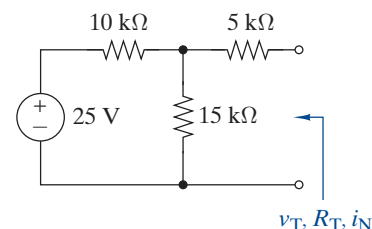


FIGURE P3-47

- 3-48 For the circuit in Figure P3-48 find its Thévenin and Norton equivalent circuits.

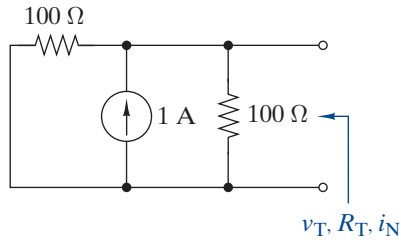


FIGURE P3-48

- 3-49 (a) Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-49.
 (b) Use the equivalent circuit found in (a) to find i_L if $R_L = 22 \text{ k}\Omega$.

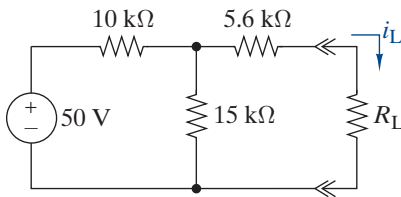


FIGURE P3-49

- 3-50 (a) Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-50.
 (b) Use the equivalent circuit found in (a) to find i_L in terms of i_S , R_1 , R_2 , and R_L .
 (c) Check your answer in (b) using current division.

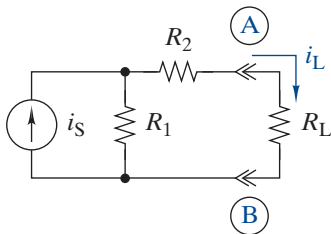


FIGURE P3-50

- 3-51 Find the Thévenin equivalent circuit seen by R_L in Figure P3-51. Find the voltage across the load when $R_L = 5 \Omega$, 10Ω , and 20Ω .

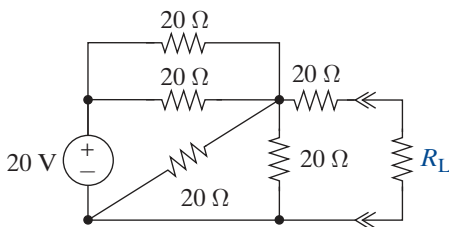


FIGURE P3-51

- 3-52 Find the Norton equivalent seen by R_L in Figure P3-52. Find the current through the load for when $R_L = 4.7 \text{ k}\Omega$, $15 \text{ k}\Omega$, and $68 \text{ k}\Omega$.

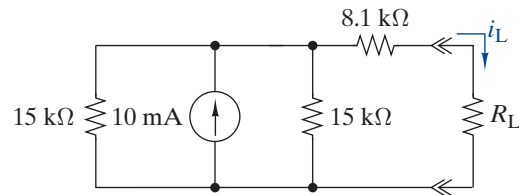


FIGURE P3-52

- 3-53 You need to determine the Thévenin Equivalent circuit of a more complex linear circuit. A technician tells you she made two measurements using her DMM. The first was with a $10\text{-k}\Omega$ load and the load current was $91 \mu\text{A}$, the second was with a $10\text{-k}\Omega$ load and the load voltage was 5 V . Calculate the Thévenin equivalent circuit as shown in Figure P3-53.

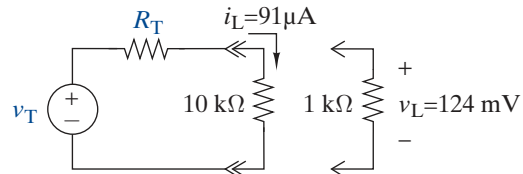


FIGURE P3-53

- 3-54 Find the Thévenin equivalent seen by R_L in Figure P3-54. Find the power delivered to the load when $R_L = 50 \text{ k}\Omega$ and $200 \text{ k}\Omega$.

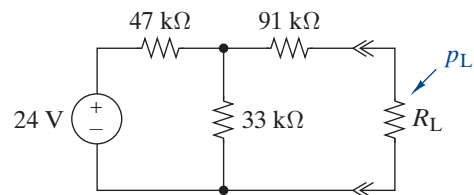


FIGURE P3-54

- 3-55 (a) Use OrCAD to find the Norton equivalent at terminals A and B in Figure P3-55. *Hint:* Find the open-circuit voltage and short-circuit current at the requisite terminals.
 (b) Use the Norton equivalent circuit found in (a) to determine the power dissipated in R_L when it is equal to $37 \text{ k}\Omega$.
 (c) Use OrCAD to simulate both the original and the Norton equivalent circuits with $R_L = 37 \text{ k}\Omega$. Verify that the power dissipated by the load is the same in both situations.

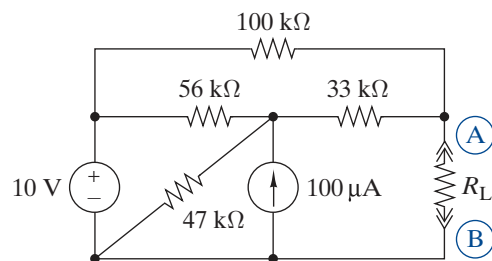


FIGURE P3-55

- 3-56** The purpose of this problem is to use Thévenin equivalent circuits to find the voltage v_X in Figure P3-56. Find the Thévenin equivalent circuit seen looking to the left of terminals A and B. Find the Thévenin equivalent circuit seen looking to the right of terminals A and B. Connect these equivalent circuits together and find the voltage v_X .

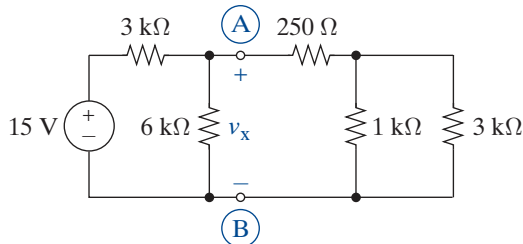


FIGURE P3-56

- 3-57** The circuit in Figure P3-57 was solved earlier using supermeshes (Prob.(3-19)). In this problem solve for the voltage across R_L by first finding the Thévenin equivalent that the load resistor sees, the for $R_L = 2.5 \text{ k}\Omega$ find v_L .

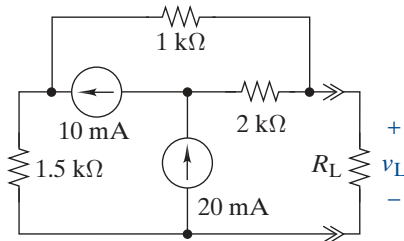


FIGURE P3-57

- 3-58** Assume that Figure P3-58 represents a model of the auxiliary output port of a car. The output current is $i = 1 \text{ A}$ when $v = 0 \text{ V}$. The output voltage is $v = 12 \text{ V}$ when a $20\text{-}\Omega$ resistor is connected between the terminals. Suppose you wanted to charge a 12-V battery by connecting the battery at the port. How much current would the port deliver to the battery?

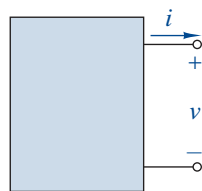


FIGURE P3-58

- 3-59** The i - v characteristic of the active circuit in Figure P3-58 is $5v + 500i = 100$. Find the output voltage when a $500\text{-}\Omega$ resistive load is connected.
- 3-60** You successfully completed Circuits I and as an undergraduate work-study, your past professor asked you to help her grade a Circuits I quiz. On the quiz, students were asked to find the power supplied by the source to both the $10 \text{ k}\Omega$ load (R_L)

and to the entire circuit as shown in Figure P3-60. Your professor asks to help her by creating a grading sheet.

- (a) Solve the quiz and establish reasonable A, B . . . etc. cuts for incorrect solutions.
- (b) A particular student correctly finds the Thévenin equivalent circuit that the resistive load sees and calculates the power to the load using V_T^2/R_L . He then does a source transformation correctly finding the Norton equivalent of the circuit. He calculates the source power by $V_T \times I_N$. What grade would you give him?
- (c) Another student finds $P_L = 5.625 \text{ mW}$ and $P_S = -22.5 \text{ mW}$, but provides no work to justify her answers. What grade would you give her?
- (d) A third student first finds the Norton equivalent, finds the current through the load using a current divider and calculates the power in the load using $I_L^2 R_L$. He figures correctly what the parallel voltage would be across the Norton circuit and the load v_L , and then calculates $P_s = I_N \times v_L$. What grade would you give him?

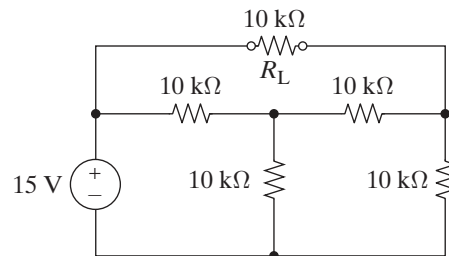


FIGURE P3-60

- 3-61** The Thévenin equivalent parameters of a practical voltage source are $v_T = 30 \text{ V}$ $R_T = 300 \text{ }\Omega$. Find the smallest load resistance for which the load voltage exceeds 10 V .
- 3-62** Use a sequence of source transformations to find the Thévenin equivalent at terminals A and B in Figure P3-62. Then select a resistor to connect across A and B so that 5 V appears across it.

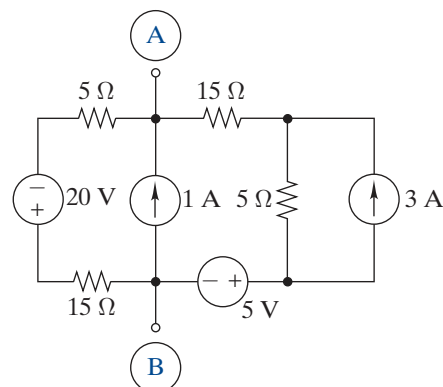


FIGURE P3-62

- 3-63** The circuit in Figure P3-63 provides power to a number of loads connected in parallel. The circuit is protected by a $\frac{3}{4} \text{ mA}$ fuse with a nominal $100 \text{ }\Omega$ resistance. Each load is $10 \text{ k}\Omega$.

What is the maximum number of loads the circuit can drive without blowing the fuse?

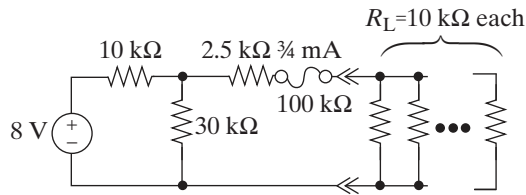


FIGURE P3-63

3-64 Find the Thévenin equivalent at terminals A and B in Figure P3-64.

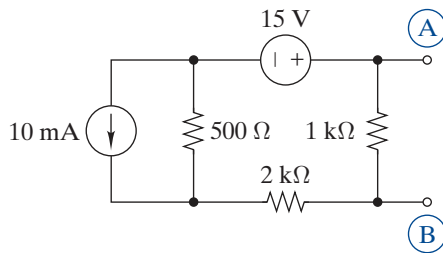


FIGURE P3-64

3-65 A nonlinear resistor is connected across a two-terminal source whose Thévenin equivalent is $v_T = 5$ V and $R_T = 500 \Omega$. The i - v characteristic of the resistor is $i = 10^{-4}(v + 2v^{3.3})$. Use the MATLAB function `solve` to find the operating point for this circuit and determine the voltage across, the current through and the power dissipated in the non-linear resistor.

3-66 A blue Light-emitting diode (LED) is connected across a two-terminal source whose Thévenin equivalent is $v_T = 4$ V and $R_T = 20 \Omega$. The i - v characteristic of the LED is $i = 10^{-12}(e^{10v} - 1)$. Figure P3-66 shows the LED's i - v characteristic. Either using MATLAB or a graphical approach determine the voltage across and current through the LED.

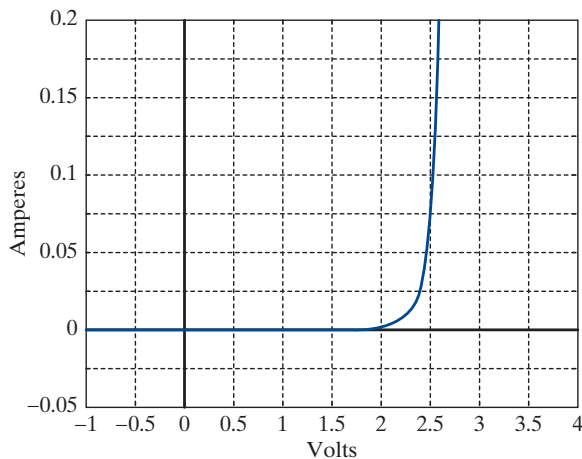


FIGURE P3-66

3-67 Find the Norton equivalent seen by R_L in Figure P3-67. Select the value of R_L so that:

- (a) 150 mA is delivered to the load.
- (b) 6 V is delivered to the load.
- (c) 100 mW is delivered to the load.

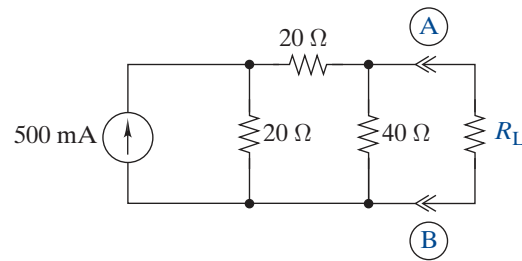


FIGURE P3-67

3-68 Find the Thévenin equivalent seen by R_L in Figure P3-68.

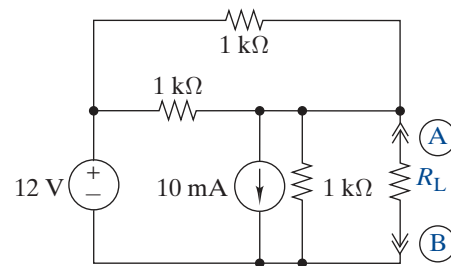


FIGURE P3-68

3-69 Find the Thévenin equivalent seen by R_L in Figure P3-69.

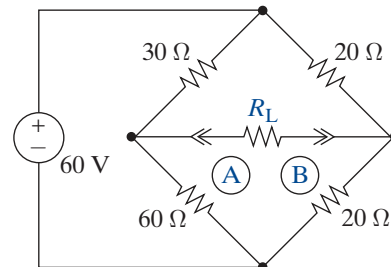


FIGURE P3-69

OBJECTIVE 3-4 MAXIMUM SIGNAL TRANSFER (SECT. 3-5)

Given a circuit containing linear resistors and independent sources,

- (a) Find the maximum voltage, current, and power available at a specified pair of terminals.
- (b) Find the resistive loads required to obtain the maximum available signal levels.
- (c) See Example 3-22 and Exercise 3-33.

3-70 For the circuit of Figure P3-70 find the value of R_L that will result in:

- (a) Maximum voltage. What is that voltage?
- (b) Maximum current. What is that current?
- (c) Maximum power. What is that power?

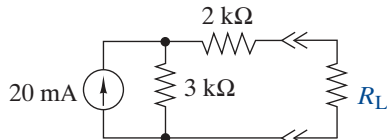


FIGURE P3-70

3-71 For the circuit of Figure P3-71 find the value of R_L that will result in:

- (d) Maximum voltage. What is that voltage?
- (e) Maximum current. What is that current?
- (f) Maximum power. What is that power?

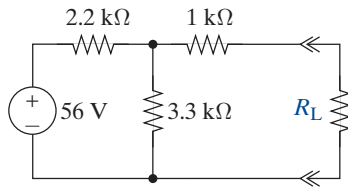


FIGURE P3-71

3-72 The resistance R in Figure P3-72 is adjusted until maximum power is delivered to the load consisting of R and the 15-kΩ resistor in parallel.

- (a) Find the required value of R .
- (b) How much power is delivered to the load?

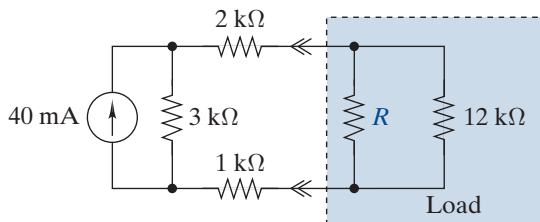


FIGURE P3-72

3-73 When a 5-kΩ resistor is connected across a two-terminal source a current of 15 mA is delivered to the load. When a second 5-kΩ resistor is connected in parallel with the first, a total current of 20 mA is delivered. Find the maximum power available from the source.

3-74 Find the value of R in the circuit of Figure P3-74 so that maximum power is delivered to the load. What is the value of the maximum power?

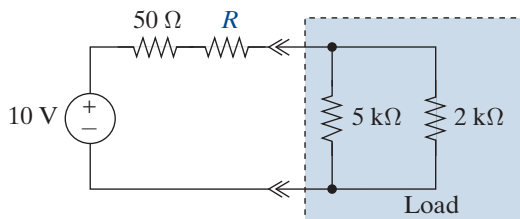


FIGURE P3-74

3-75 For the circuit of Figure P3-75 find the value of R_L that will result in:

- (g) Maximum voltage. What is that voltage?
- (h) Maximum current. What is that current?
- (i) Maximum power. What is that power?

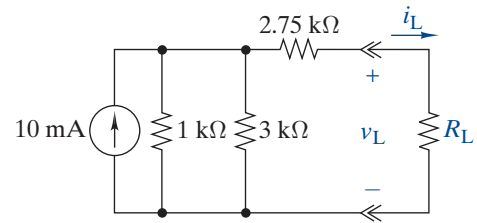


FIGURE P3-75

3-76 **D** A 100 Ω-load needs 10 mA to operate correctly. Design a practical power source to provide the needed current. The smallest source resistance you can practically design for is 50 Ω, but you can add any other series resistance if you need to.

3-77 A practical source delivers 50 mA to a 300-Ω load. The source delivers 12 V to a 120-Ω load. Find the maximum power available from the source.

OBJECTIVE 3-5 INTERFACE CIRCUIT DESIGN AND EVALUATION (SECT. 3-6)

- (a) **D** Given the signal transfer objective at a source-load interface, adjust the circuit parameters or design one or more two-port interface circuits to achieve the specified objectives within stated constraints.
- (b) **E** Given two or more circuits that perform the same interface function, rank-order the circuits using stated criteria.

See Examples 3-23, 3-24, 3-25, 3-26, 3-27, 3-28, 3-29, 3-30, 3-31 and Exercises 3-34, 3-35, 3-36, 3-37, 3-38, 3-39, 3-40, 3-41, 3-42, 3-43.

3-78 **D** Select R_L and design an interface circuit for the circuit shown in Figure P3-78 so that the load voltage is 2 V.

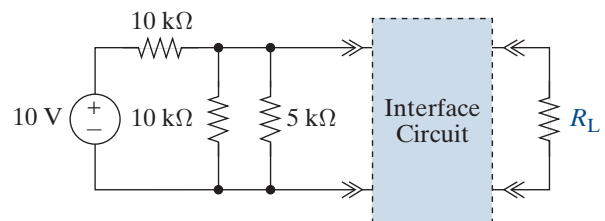


FIGURE P3-78

3-79 **D** The source in Figure P3-79 has a 100 mA output current limit. Design an interface circuit so that the load voltage is $v_2 = 20$ V and the source current is $i_1 < 50$ mA.

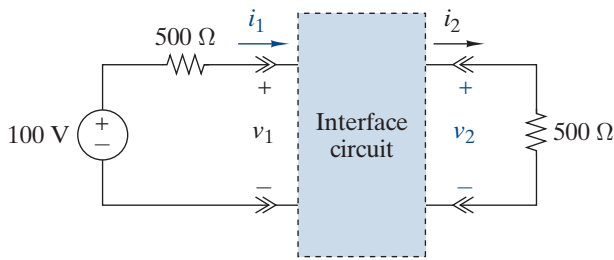


FIGURE P3-79

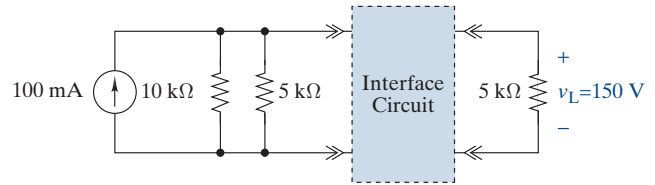


FIGURE P3-82

- 3-80 D** Figure P3-80 shows an interface circuit connecting a 15-V source to a diode load. The i - v characteristic of the diode is $i = 10^{-14} (e^{40v} - 1)$.
- Design an interface circuit so that $v = 0.7$ V.
 - Validate your answer using MATLAB.

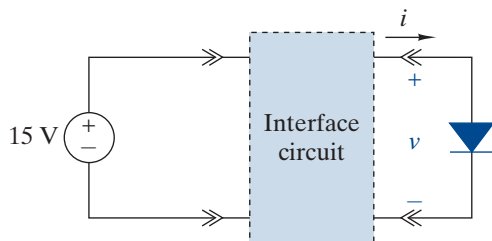


FIGURE P3-80

- 3-81 D** Design the interface circuit in Figure P3-81 so that the voltage delivered to the load is $v = 10$ V $\pm 10\%$. Use only one or more of the following standard resistors: 1.3 k Ω , 2 k Ω , 3 k Ω , 4.3 k Ω , 6.2 k Ω , and 9.1 k Ω . These resistors all have a tolerance of $\pm 5\%$ which you must account for in your design.

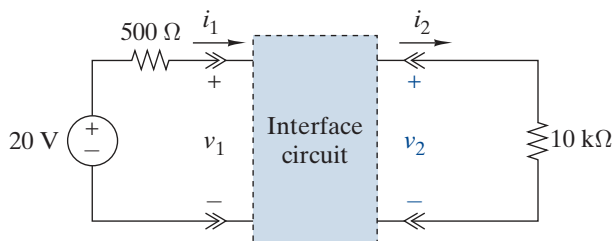
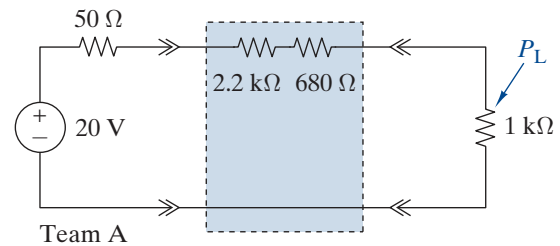


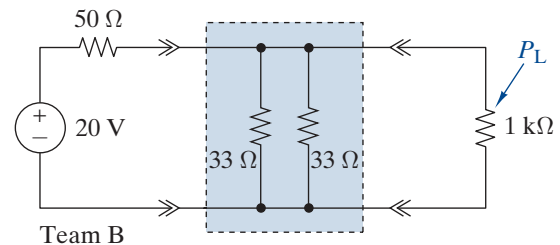
FIGURE P3-81

- 3-82 D, E** In this problem you will design two interface circuits that deliver 150 V to the 5-k Ω load.
- Design a parallel resistor interface to meet the requirements.
 - Convert the source circuit to its Thévenin equivalent and then design a series interface to meet the requirement.
 - If minimizing the power that the current source delivers is the primary consideration, which of your two designs best meets the requirement?

- 3-83 E** Two teams are competing to design the interface circuit in Figure P3-83 so that the 25 mW $\pm 10\%$ is delivered to the 1-k Ω load resistor. Their designs are shown in Figure P3-83. Which solution is better considering the use of standard values, number of parts, and power required by the source? Would your choice be different if the power had to be within $\pm 5\%$?



Team A



Team B

FIGURE P3-83

- 3-84 E** The bridge-T attenuation pad shown in Figure P3-84 was found in a drawer. You need an attenuation pad that would match to a 75- Ω source and a 75- Ω load and provide for a -12 dB drop of signal (reduction of four times). Use OrCAD to determine if the device will work.

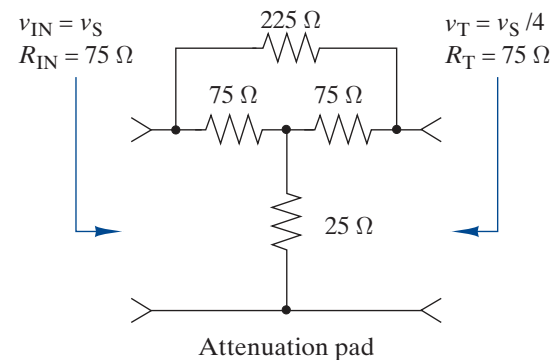


FIGURE P3-84

- 3-85 **D** Design the interface circuit in Figure P3-85 so that the power delivered to the load is 100 mW.

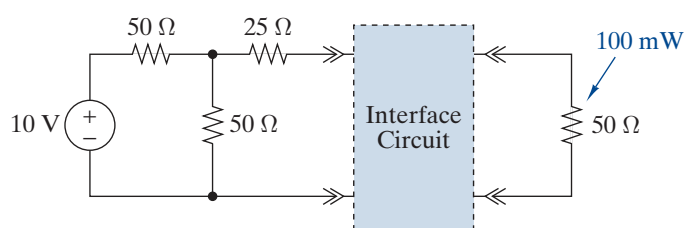


FIGURE P3-85

- 3-86 **D** Design the interface circuit in Figure P3-85 so that the voltage delivered to the load is 2.5 V.

- 3-87 **D** Design the interface circuit in Figure P3-87 so that $R_{IN} = 100 \Omega$ and the current delivered to the 50- Ω load is $i = 50$ mA. Hint: Use an L-pad.

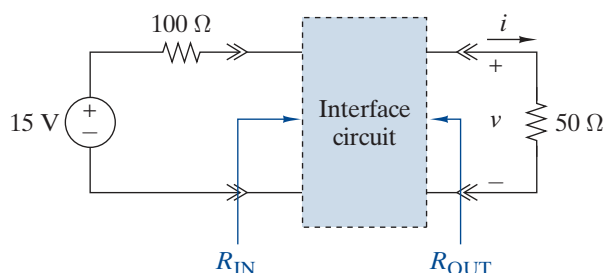


FIGURE P3-87

- 3-88 **D** Design the interface circuit in Figure P3-87 so that $R_{OUT} = 50 \Omega$ and the voltage delivered to the 50- Ω load is $v = 2.5$ V. Hint: Use an L-pad.

- 3-89 **D** The circuit in Figure P3-89 has a source resistance of 75 Ω and a load resistance of 300 Ω . Design the interface circuit so that the input resistance is $R_{IN} = 75 \Omega \pm 10\%$ and the output resistance is $R_{OUT} = 300 \Omega \pm 10\%$.

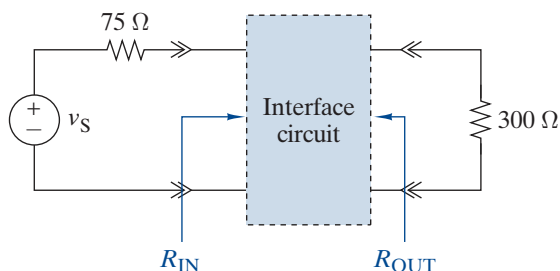


FIGURE P3-89

- 3-90 **E** It is claimed that both interface circuits in Figure P3-90 will deliver $v = 4$ V to the 75- Ω load. Verify this claim. Which interface circuit consumes the least power? Which has an output resistance that best matches the 75- Ω load?

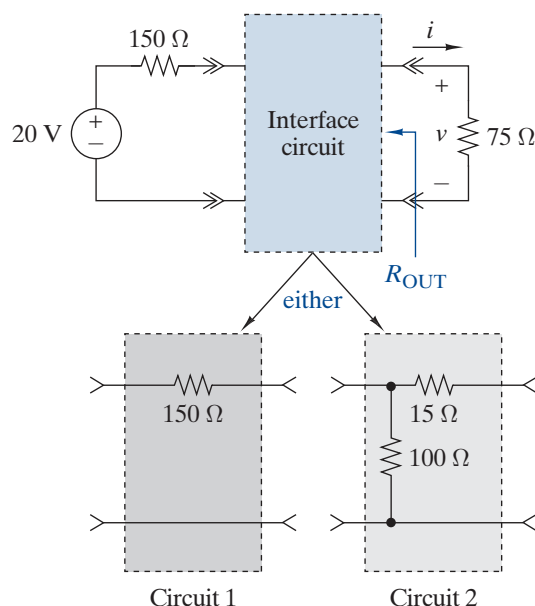


FIGURE P3-90

INTEGRATING PROBLEMS

- 3-91 **E** Audio Speaker Resistance-Matchign Network

A company is producing an interface network that they claim would result in an R_{IN} of $600 \Omega \pm 2\%$ and R_{OUT} of 16, 8, or $4 \Omega \pm 2\%$ – depending on whether the connected speakers are 16, 8, or 4Ω – selectable via a built-in switch. Their design is shown in Figure P3-91. Prove or disprove their claim.

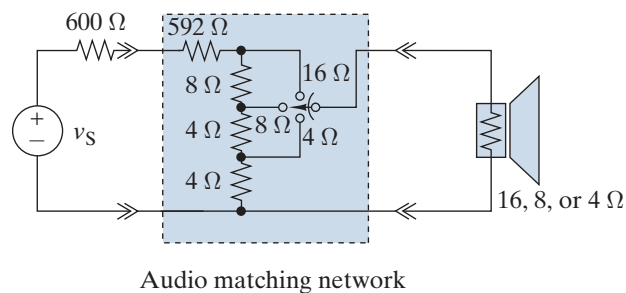


FIGURE P3-91

- 3-92 **A** Attenuation Analysis

In Figure P3-92 a two-port attenuator connects a 600- Ω source to a 600- Ω load. Find the power delivered to the load in terms of v_S . Remove the attenuator and find the power delivered to the load when the source is directly connected to the load. By what fraction does the attenuator reduce the power delivered to the 600- Ω load? Express the fraction in dB.

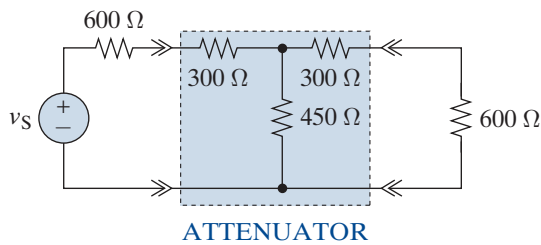


FIGURE P3-92

3-93 **D** Interface Circuit Design

Using no more than three $50\text{-}\Omega$ resistors, design the interface circuit in Figure P3-93 so that $v \leq 5\text{ V}$ and $i \leq 100\text{ mA}$ regardless of the value of R_L .

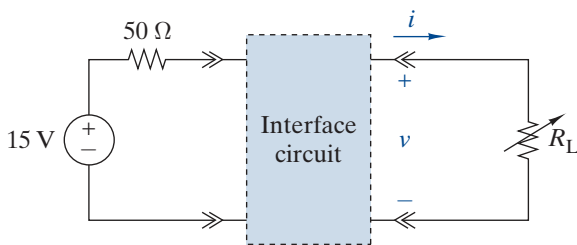


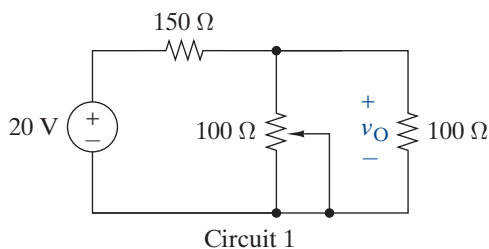
FIGURE P3-93

3-94 **D** Battery Design

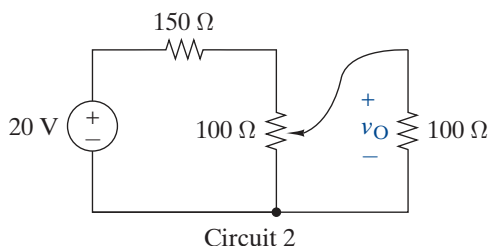
A satellite requires a battery with an open-circuit voltage $v_{OC} = 36\text{ V}$ and a Thévenin resistance $R_T \leq 10\text{ }\Omega$. The battery is to be constructed using series and parallel combinations of one of two types of cells. The first type has $v_{OC} = 9\text{ V}$, $R_T = 4\text{ }\Omega$, and a weight of 30 grams. The second type has $v_{OC} = 4\text{ V}$, $R_T = 2\text{ }\Omega$, and a weight of 18 grams. Design a minimum weight battery that meets the open-circuit voltage and Thévenin resistance requirements.

3-95 **E** Design Evaluation

A requirement exists for a circuit to deliver 0 to 5 V to a $100\text{-}\Omega$ load from a 20-V source rated at 2.5 W. Two proposed circuits are shown in Figure P3-95. Which one would you choose and why?



Circuit 1



Circuit 2

FIGURE P3-95

3-96 **E** Design Interface Competition.

The output of a transistorized power supply is modeled by the Norton equivalent circuit shown Figure P3-96. Two teams are competing to design the interface circuit so that the $25\text{ mW} \pm 10\%$ is delivered to the $1\text{-k}\Omega$ load resistor. Their designs are shown in Figure P3-96. Which solution is better considering the use of standard values, number of parts, and power required by the source? Would your choice be different if the power had to be within $\pm 5\%$?

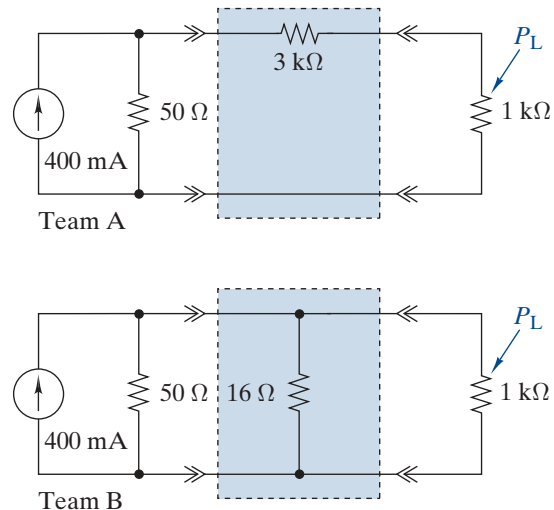
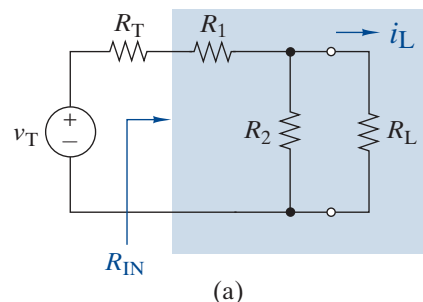


FIGURE P3-96

3-97 **A, E** Analysis of Competing Interface Circuits Using MATLAB.

Figure P3-97 displays two generalized interface circuit designs. In both circuits, resistors R_1 and R_2 connect a Thévenin equivalent circuit to a load resistor. Using MATLAB, develop symbolic expressions for the load resistance, i_L , and the input resistance, R_{IN} , for each circuit in terms of the given parameters. Using these two expressions, now use the MATLAB command `solve` to solve for R_1 and R_2 in terms of i_L and R_{IN} . Let $v_T = 15\text{ V}$, $R_T = 100\text{ }\Omega$, $R_L = 50\text{ }\Omega$, $i_L = 50\text{ mA}$, and $R_{IN} = 100\text{ }\Omega$. Can you use both types of interface circuits to find suitable values for R_1 and R_2 to meet these specifications? Compare your interface design(s) with the solution to Problem 3-87.



(a)

FIGURE 3-97

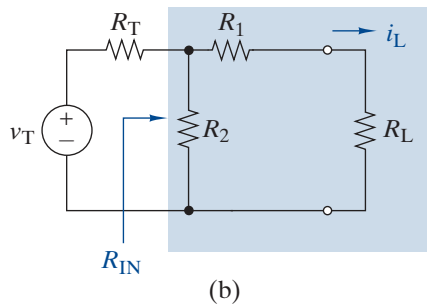


FIGURE P3-97 (Continued)

3-98 Maximum Power Transfer Using OrCAD

Figure P3-98 shows a circuit with two sources, a fixed load and a resistor R . Select R for maximum power transfer to the load. The result is not an obvious one. Hint: Simulate in OrCAD using the PARAMETER function and sweep R from $0.1\ \Omega$ to $100\ \text{M}\Omega$.

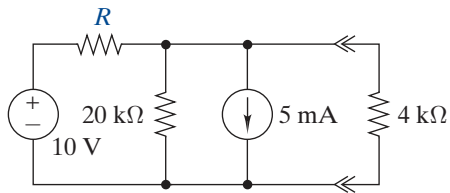


FIGURE P3-98

3-99 Non-inverting Summer

A *non-inverting summer* interface device is shown in Figure P3-99. Of importance is that the input to the device has infinite resistance – i.e. no current flows into the device. The output voltage for this configuration is two times whatever the input voltage is. Develop a relationship for the voltage v_L across R_L with respect to the two input voltages and the two input resistances.

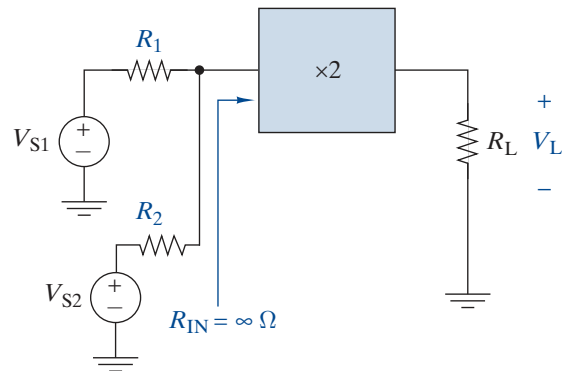


FIGURE P3-99